

Continuation of Polar Co-ordinates

Recap

A parametric equation takes the form:

$$x \equiv x(\theta), y \equiv y(\theta) \text{ with } \theta \in \mathbb{R}$$

We regard x, y as functions of θ

e.g. $x^2 + y^2 = r^2$ (circle)

Parameter $\theta \in [0, 2\pi]$ Radius

$$x, y = r \cos \theta, r \sin \theta$$

$$(r \cos \theta)^2 + (r \sin \theta)^2$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta) = r^2 \Rightarrow \text{consistent}$$

Consider ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

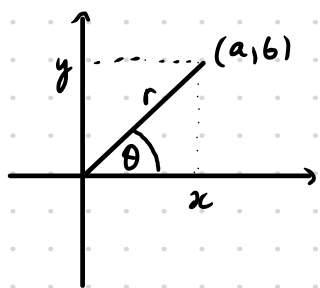
$$\rightarrow x, y = a \cos \theta, b \sin \theta$$

Parabola: $y^2 = uax$

$$\rightarrow x, y = 2at, at^2 \text{ where } t \text{ is a parameter, } t \in \mathbb{R}$$

Polar Coordinate representation

$$(x, y) \leftrightarrow (r, \theta): x, y = r \cos \theta, r \sin \theta$$



r = distance from origin

θ = angle measured from positive x -axis

$\theta \rightarrow \theta + 2\pi$ maps to the same point

If r is negative, then take $|r|$ and replace θ with $\theta + \pi$
 $(-r^*, \theta^*) \rightarrow (r^*, \theta^* + \pi)$

Try to simplify to $r = f(\theta)$

Move on tangents and normals

Normal is orthogonal to tangent

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$$